

FINAL PROJECT

DYH

2025-08-21

```
# LOAD LIBRARIES AND IMPORT DATA
```

```
library(tidyverse) # Core data science suite: dplyr, tidyr, ggplot2, readr etc.
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.1.4      v readr      2.1.5
```

```
## v forcats    1.0.0      v stringr   1.5.1
```

```
## v ggplot2    3.5.2      v tibble    3.3.0
```

```
## v lubridate  1.9.4      v tidyr     1.3.1
```

```
## v purrr      1.0.4
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(janitor) # Data cleaning (clean_names, etc.)
```

```
##
```

```
## Attaching package: 'janitor'
```

```
##
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
##      chisq.test, fisher.test
```

```
library(gtsummary) # Generate summary tables (supports LaTeX output)
```

```
library(kableExtra) # Enhanced table formatting (supports booktabs, striped LaTeX options)
```

```
##
```

```
## Attaching package: 'kableExtra'
```

```
##
```

```
## The following object is masked from 'package:dplyr':
```

```
##
```

```
##      group_rows
```

```
# DATA IMPORT
```

```
# Using show_col_types=FALSE suppresses column type messages
```

```
genes_data <- read_csv("D:/2025_summer/R/QBS103_GSE157103_genes.csv",  
                      show_col_types = FALSE)
```

```
## New names:
```

```
## * ' ' -> '...1'
```

```
series_matrix <- read_csv("D:/2025_summer/R/QBS103_GSE157103_series_matrix-1.csv",
                          show_col_types = FALSE)
```

```
# DATA CLEANING AND PREPARATION
series_matrix_clean <- series_matrix %>%
  clean_names() %>%
  mutate(
    ## Age processing
    age_years = case_when(
      !is.na(age) & str_detect(age, ">\\s*89") ~ 90, # Handle ">89" cases
      TRUE ~ suppressWarnings(as.numeric(str_extract(age, "\\d+")))
    ),
    ## Gender standardization
    sex = str_to_title(str_trim(sex)),
    ## ICU status factor
    icu_status = factor(icu_status,
                        levels = c("no", "yes"),
                        labels = c("Non-ICU", "ICU")),
    ## CRP cleaning
    crp_mg_l = na_if(crp_mg_l, "unknown"),
    crp_mg_l = readr::parse_number(as.character(crp_mg_l)),
    ## Disease classification
    disease_status = if_else(str_detect(disease_status, "COVID-19"),
                             "COVID-19", "Non-COVID-19")
  )

# Variable classification
cont_vars <- c("age_years", "crp_mg_l") # Continuous variables
cat_vars <- c("sex", "disease_status") # Categorical variables

# Prepare table data
tab_df <- series_matrix_clean %>%
  select(icu_status, all_of(cont_vars), all_of(cat_vars))

# GENERATE SUMMARY TABLE
tbl <- tbl_summary(
  data = tab_df,
  by = icu_status, # Stratify by ICU status
  statistic = list(
    all_categorical() ~ "{n} ({p}%)", # Categorical format
    age_years ~ "{mean} ({sd}); {median} [{p25}, {p75}]", # Continuous format
    crp_mg_l ~ "{mean} ({sd}); {median} [{p25}, {p75}]"
  ),
  label = list( # Improved labels with type indication
    age_years ~ "Age (years) (Continuous)",
    crp_mg_l ~ "CRP (mg/L) (Continuous)",
    sex ~ "Sex (Categorical)",
    disease_status ~ "Disease status (Categorical)"
  ),
  missing_text = "Missing" # Explicit missing data handling
) %>%
  add_n() %>% # Add sample size column
  add_p(test = list( # Statistical tests
```

Variable	N	Non-ICU N = 60 ¹	ICU N = 66 ¹	p-value ²
Age (years) (Continuous)	125	60 (18); 58 [49, 77]	63 (14); 65 [55, 74]	0.2
Missing		1	0	
CRP (mg/L) (Continuous)	107	109 (94); 109 [38, 146]	150 (106); 136 [51, 235]	0.035
Missing		11	8	
Sex (Categorical)	126			0.4
Female		27 (45%)	24 (36%)	
Male		33 (55%)	41 (62%)	
Unknown		0 (0%)	1 (1.5%)	
Disease status (Categorical)	126			
COVID-19		60 (100%)	66 (100%)	

¹Mean (SD); Median [Q1, Q3]; n (%)

²Wilcoxon rank sum test; Pearson's Chi-squared test; NA

```
all_categorical() ~ "chisq.test",
all_continuous() ~ "wilcox.test"
)) %>%
modify_header(label ~ "**Variable**") %>% # Format header
bold_labels() # Emphasize labels
```

```
## The following errors were returned during 'modify_header()':
## x For variable 'disease_status' ('icu_status') and "statistic", "p.value", and
## "parameter" statistics: 'x' and 'y' must have at least 2 levels
## The following warnings were returned during 'modify_header()':
## ! For variable 'sex' ('icu_status') and "statistic", "p.value", and "parameter"
## statistics: Chi-squared approximation may be incorrect
```

```
# PREVIEW IN RSTUDIO
tbl # Display HTML table in Viewer
```

```
# LATEX TABLE OUTPUT
tbl %>%
as_kable_extra(
  format = "latex",
  booktabs = TRUE, # Professional table formatting
  linesep = "", # Remove extra line spacing
  latex_options = c("hold_position", "striped") # Table positioning and styling
)
```

Variable	N	Non-ICU N = 60	ICU N = 66	p-value
Age (years) (Continuous)	125	60 (18); 58 [49, 77]	63 (14); 65 [55, 74]	0.2
Missing		1	0	
CRP (mg/L) (Continuous)	107	109 (94); 109 [38, 146]	150 (106); 136 [51, 235]	0.035
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¹ Mean (SD); Median [Q1, Q3]; n (%)

² Wilcoxon rank sum test; Pearson's Chi-squared test; NA

Publication Quality Visualizations

```
# PUBLICATION-QUALITY PLOTS
library(tidyverse)
library(RColorBrewer)

# REIMPORT DATA FOR FRESH START
genes_data <- read_csv("D:/2025_summer/R/QBS103_GSE157103_genes.csv",
  show_col_types = FALSE)

## New names:
## * ' ' -> '...1'

series_matrix <- read_csv("D:/2025_summer/R/QBS103_GSE157103_series_matrix-1.csv",
  show_col_types = FALSE)

# RESHAPE GENE DATA TO LONG FORMAT
expr_long <- genes_data %>%
  pivot_longer(
    cols = -1,
    names_to = "participant_id",
    values_to = "expression"
  ) %>%
  rename(gene = 1)

# CLEAN METADATA
meta_df <- series_matrix %>%
  rename(crp = `crp(mg/l)`) %>%
  mutate(
    crp = parse_number(crp),
    sex = factor(str_trim(sex)),
    icu_status = factor(str_trim(icu_status),
      levels = c("no", "yes"),
      labels = c("No ICU", "ICU"))
  ) %>%
  select(participant_id, crp, sex, icu_status)
```

```
## Warning: There was 1 warning in 'mutate()'.
## i In argument: 'crp = parse_number(crp)'.
## Caused by warning:
## ! 19 parsing failures.
## row col expected actual
## 2 -- a number unknown
## 7 -- a number unknown
## 48 -- a number unknown
## 52 -- a number unknown
## 58 -- a number unknown
## ... ..
## See problems(...) for more details.
```

PLOTTING FUNCTION DEFINITION

```
plot_gene_plots <- function(data, gene_list,
                             cont_var, cat_var1, cat_var2) {
  # Filter and clean data
  df <- data %>%
    filter(gene %in% gene_list) %>%
    drop_na(all_of(c(cont_var, cat_var1, cat_var2)))

  if (nrow(df) == 0) stop("No data to plot: check gene names or covariates.")

  # Generate plots for each gene
  for (g in gene_list) {
    sub <- df %>% filter(gene == g)

    # 1. HISTOGRAM WITH DENSITY
    p_hist <- ggplot(sub, aes(x = expression)) +
      geom_histogram(aes(y = ..density..),
                     bins = 30, fill = "#2c7fb8",
                     color = "white", alpha = 0.85) +
      geom_density(color = "#08306b", size = 1, alpha = 0.7) +
      geom_vline(aes(xintercept = mean(expression, na.rm = TRUE)),
                 color = "red", linetype = "dashed", size = 1) +
      annotate("text",
              x = mean(sub$expression, na.rm = TRUE),
              y = 0,
              label = paste0("Mean = ",
                             round(mean(sub$expression, na.rm = TRUE), 2)),
              vjust = -1, hjust = 0, color = "red", size = 4) +
      labs(title = paste("Distribution of", g, "Expression"),
           x = paste(g, "Expression"),
           y = "Density") +
      theme_classic(base_size = 14) +
      theme(
        plot.title = element_text(face = "bold", hjust = 0.5),
        axis.text = element_text(color = "black")
      )
    print(p_hist)

    # 2. SCATTERPLOT WITH TRENDLINE
    p_scatter <- ggplot(sub,
                        aes_string(x = cont_var,
```

```

        y      = "expression",
        color = cat_var1,
        shape = cat_var2)) +
geom_point(size = 2.8, alpha = 0.75) +
geom_smooth(method = "lm", se = FALSE,
            linetype = "dashed", color = "gray30", size = 0.8) +
labs(
  title = paste0(g, " Expression vs ", toupper(cont_var)),
  subtitle = paste("Colored by", tools::toTitleCase(cat_var1),
                  "and shaped by ICU Status"),
  x      = paste0(tools::toTitleCase(cont_var), " (mg/L)"),
  y      = paste(g, "Expression"),
  color = tools::toTitleCase(cat_var1),
  shape = "ICU Status"
) +
scale_color_brewer(palette = "Set1") + # Colorblind-friendly palette
scale_shape_manual(values = c(16, 17)) + # Distinct shapes
theme_bw(base_size = 14) +
theme(
  plot.title      = element_text(face = "bold", hjust = 0.5, size = 16),
  plot.subtitle   = element_text(hjust = 0.5, size = 12, color = "gray30"),
  legend.position = "right"
)
print(p_scatter)

# 3. BOXPLOT WITH JITTER
p_box <- ggplot(sub,
               aes_string(x = cat_var1,
                          y = "expression",
                          fill = cat_var2)) +
geom_boxplot(outlier.size = 1, alpha = 0.85) +
geom_jitter(width = 0.15, alpha = 0.5, size = 1) +
scale_fill_brewer(palette = "Set2", name = "ICU Status") +
labs(
  title = paste0(g, " Expression by ", tools::toTitleCase(cat_var1),
                  " and ICU Status"),
  subtitle = "Boxplots show median & IQR; points show individual samples",
  x      = tools::toTitleCase(cat_var1),
  y      = paste(g, "Expression")
) +
theme_minimal(base_size = 14) +
theme(
  plot.title      = element_text(face = "bold", hjust = 0.5, size = 16),
  plot.subtitle   = element_text(hjust = 0.5, size = 12, color = "gray30"),
  legend.position = "top",
  legend.title    = element_text(face = "bold")
)
print(p_box)
}
}

# MERGE EXPRESSION AND METADATA
merged_df <- expr_long %>%

```

```

left_join(meta_df, by = "participant_id")

# GENERATE ABCA1 PLOTS
plot_gene_plots(
  data      = merged_df,
  gene_list = c("ABCA1"), # Focus on ABCA1 gene
  cont_var  = "crp",       # Continuous variable
  cat_var1  = "sex",       # Primary categorical
  cat_var2  = "icu_status" # Secondary categorical
)

```

```

## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.

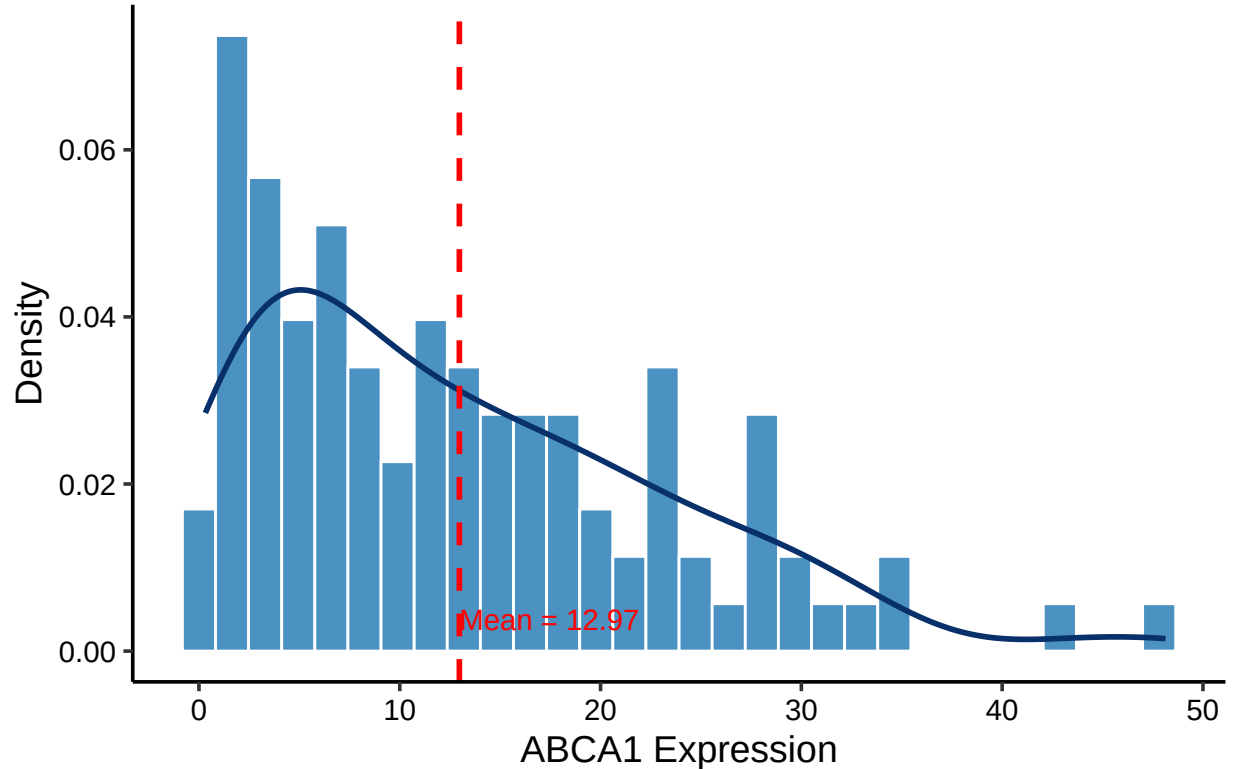
```

```

## Warning: The dot-dot notation ('..density..') was deprecated in ggplot2 3.4.0.
## i Please use 'after_stat(density)' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.

```

Distribution of ABCA1 Expression



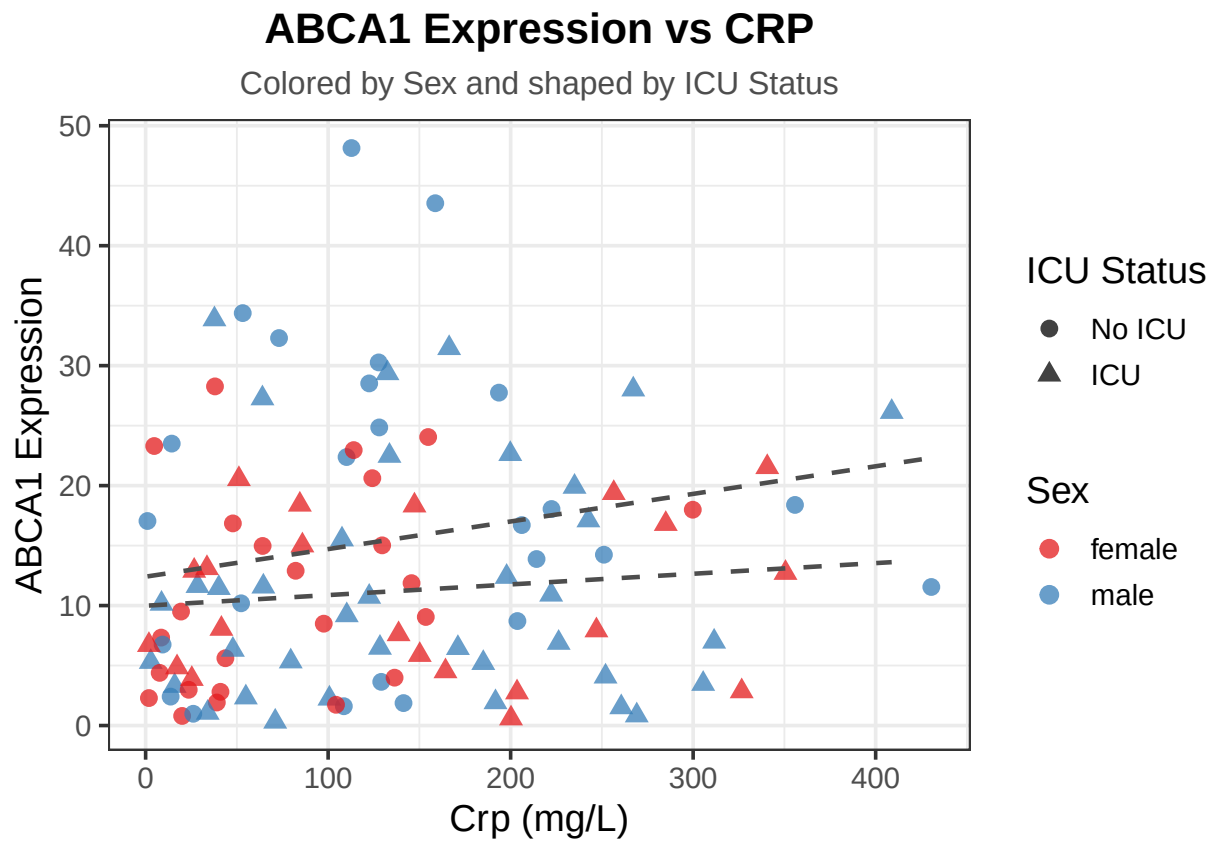
```

## Warning: 'aes_string()' was deprecated in ggplot2 3.0.0.

```

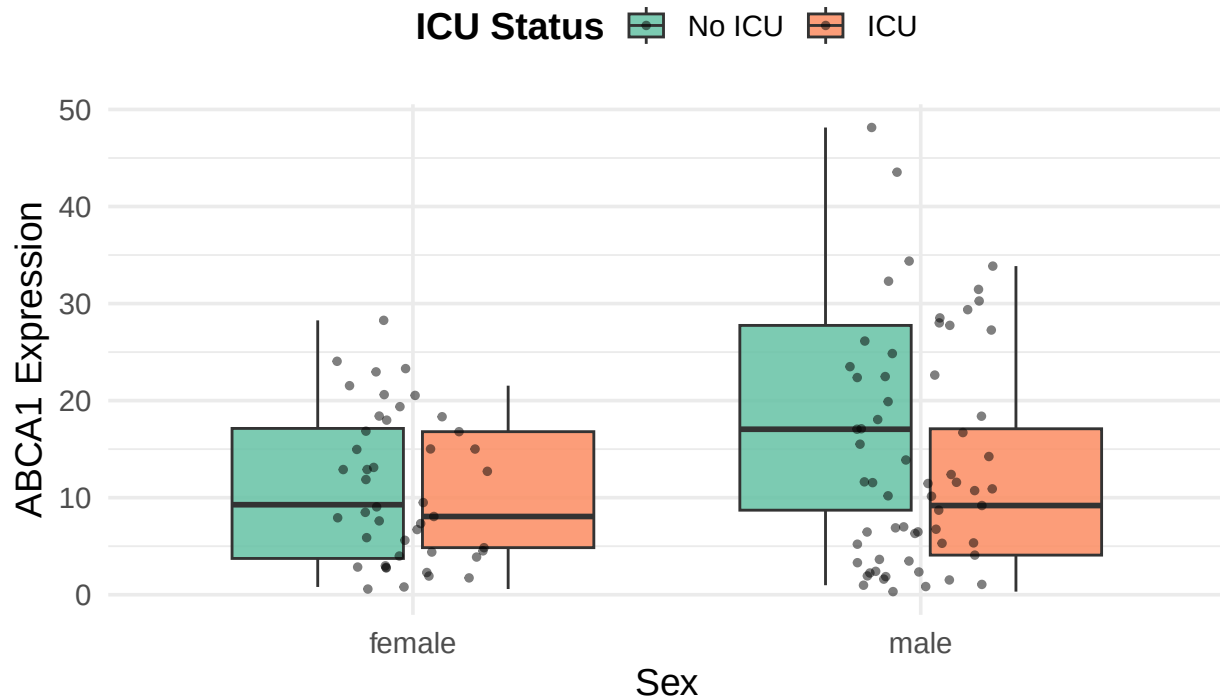
```
## i Please use tidy evaluation idioms with 'aes()'.
## i See also 'vignette("ggplot2-in-packages")' for more information.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



ABCA1 Expression by Sex and ICU Status

Boxplots show median & IQR; points show individual samples



Heatmap Visualization

```
# HEATMAP OF TOP VARIABLE GENES
library(dplyr)
library(pheatmap)
library(RColorBrewer)

# 1. DATA IMPORT
genes_df <- read.csv("QBS103_GSE157103_genes.csv", check.names = FALSE, row.names = 1)
series_df <- read.csv("QBS103_GSE157103_series_matrix-1.csv", check.names = FALSE)

# 2. EXPRESSION MATRIX PREPARATION
expr_mat <- as.data.frame(t(genes_df)) # Transpose to samples x genes
expr_mat <- mutate_all(expr_mat, function(x) as.numeric(as.character(x)))

# Sample alignment
common_ids <- intersect(rownames(expr_mat), series_df$participant_id)
expr_mat <- expr_mat[common_ids, , drop = FALSE]
series_df <- series_df[match(common_ids, series_df$participant_id), , drop = FALSE]

# 3. METADATA CLEANING
series_df$sex <- tolower(trimws(series_df$sex))
series_df$icu_status <- tolower(trimws(series_df$icu_status))
```

```

# Standardize values
series_df$sex <- case_when(
  series_df$sex == "male" ~ "Male",
  series_df$sex == "female" ~ "Female",
  TRUE ~ "Unknown"
)

series_df$icu_status <- ifelse(series_df$icu_status == "yes", "ICU", "Non-ICU")

# Prepare annotation data
rownames(series_df) <- series_df$participant_id
anno_row <- series_df[, c("icu_status", "sex"), drop = FALSE]

# Filter unknown genders
keep_idx <- anno_row$sex %in% c("Male", "Female") & !is.na(anno_row$sex)
expr_mat <- expr_mat[keep_idx, , drop = FALSE]
anno_row <- droplevels(anno_row[keep_idx, , drop = FALSE])

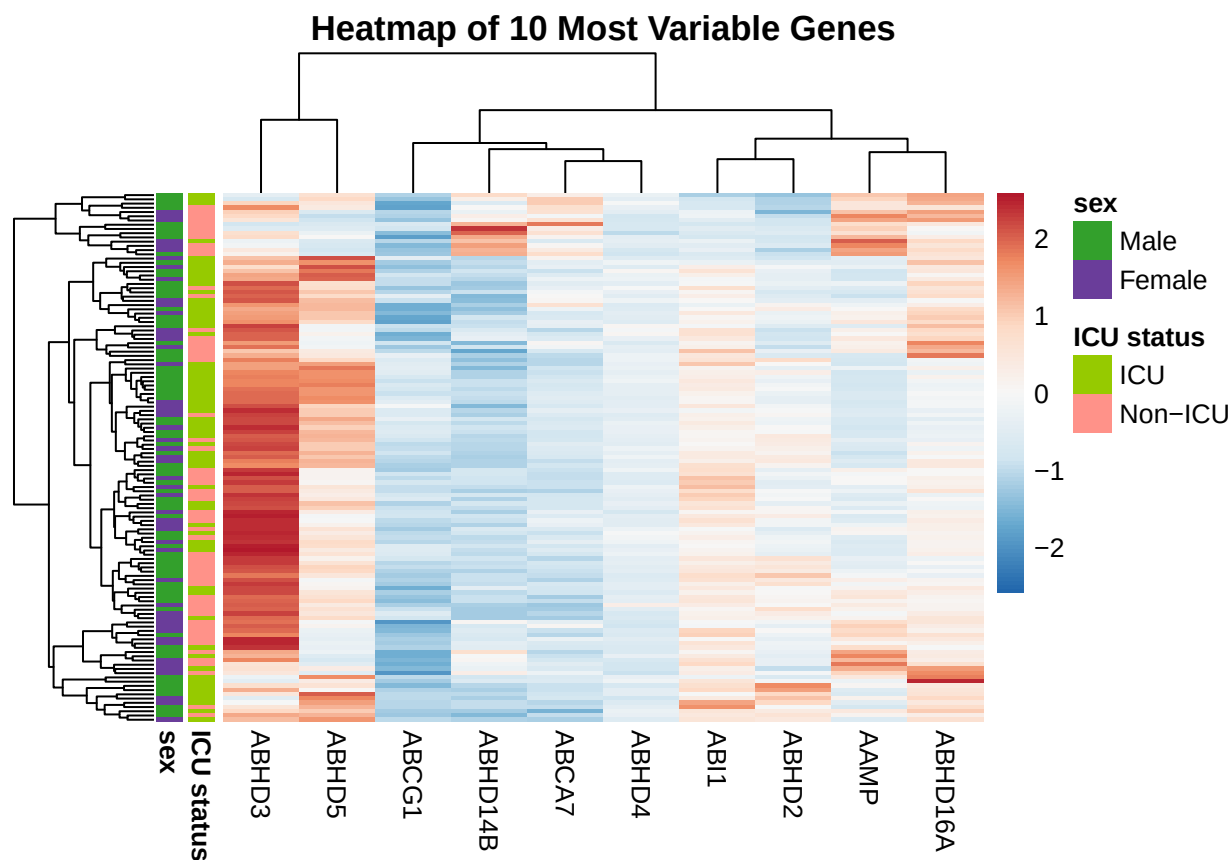
# 4. SELECT TOP VARIABLE GENES
N <- 10 # Number of top genes to display
gene_vars <- apply(expr_mat, 2, var, na.rm = TRUE)
top_genes <- names(sort(gene_vars, decreasing = TRUE))[seq_len(min(N, length(gene_vars)))]
expr_mat <- expr_mat[, top_genes, drop = FALSE]

# 5. MISSING VALUE IMPUTATION
for (i in seq_len(nrow(expr_mat))) {
  if (anyNA(expr_mat[i, ])) {
    row_mean <- mean(expr_mat[i, ], na.rm = TRUE)
    expr_mat[i, is.na(expr_mat[i, ])] <- row_mean
  }
}

# 6. HEATMAP VISUALIZATION
ann_colors <- list(
  icu_status = c("Non-ICU" = "#1f78b4", "ICU" = "#e31a1c"), # Blue/Red
  sex = c("Male" = "#33a02c", "Female" = "#6a3d9a") # Green/Purple
)
colnames(anno_row)[colnames(anno_row) == "icu_status"] <- "ICU status"

pheatmap(
  as.matrix(expr_mat),
  scale = "row", # Row-normalize (z-score)
  annotation_row = anno_row,
  annotation_colors = ann_colors,
  cluster_rows = TRUE,
  cluster_cols = TRUE,
  show_rownames = FALSE, # Hide sample names
  show_colnames = TRUE, # Show gene names
  fontsize = 10,
  border_color = NA, # No cell borders
  color = colorRampPalette(rev(brewer.pal(7, "RdBu")))(100), # Diverging colors
  main = sprintf("Heatmap of %d Most Variable Genes", N)
)

```



new plot type

```
# PUBLICATION-QUALITY ABCA1 VISUALIZATION
library(tidyverse)

# 1. DATA PREPARATION (REPEATED FOR COMPLETENESS)
expr_long <- genes_data %>%
  rename(gene = 1) %>%
  pivot_longer(-gene, names_to = "participant_id", values_to = "expression") %>%
  mutate(
    gene = toupper(trimws(gene)),
    expression = as.numeric(expression)
  )

series_matrix_clean <- series_matrix %>%
  mutate(age = parse_number(as.character(age))) %>%
  select(participant_id, age)
```

```
## Warning: There was 1 warning in 'mutate()'.
## i In argument: 'age = parse_number(as.character(age))'.
## Caused by warning:
## ! 1 parsing failure.
## row col expected actual
```

```
## 6 -- a number :
```

```
df_abca1 <- expr_long %>%
  filter(gene == "ABCA1") %>%
  left_join(series_matrix_clean, by = "participant_id") %>%
  drop_na(age, expression)

# 2. QUANTILE DATA RESHAPING
summ_age_long <- df_abca1 %>%
  group_by(age) %>%
  summarise(
    Q25 = quantile(expression, 0.25, na.rm = TRUE),
    Median = quantile(expression, 0.50, na.rm = TRUE),
    Q75 = quantile(expression, 0.75, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  pivot_longer(cols = c(Q25, Median, Q75),
    names_to = "Quantile", values_to = "Expression")

segments_df <- summ_age_long %>%
  pivot_wider(names_from = Quantile, values_from = Expression) %>%
  select(age, Q25, Q75)

# 3. AESTHETIC MAPPINGS
color_map <- c(
  "Q25" = "#0072B2",      # Blue
  "Q75" = "#D55E00",      # Orange
  "Median" = "black",     # Black
  "Median trend" = "black"
)
fill_map <- c(
  "Q25" = "#0072B2",
  "Q75" = "#D55E00",
  "Median" = "white",
  "Median trend" = NA
)
shape_map <- c(
  "Q25" = 21, # Filled circle
  "Q75" = 21,
  "Median" = 21,
  "Median trend" = NA
)

# 4. FINAL PLOT CONSTRUCTION
p_abca1 <- ggplot() +
  # Q25-Q75 segments (gray lines)
  geom_segment(data = segments_df,
    aes(x = age, xend = age, y = Q25, yend = Q75),
    color = "grey60", linewidth = 1) +
  # Quantile points
  geom_point(data = summ_age_long,
    aes(x = age, y = Expression,
      color = Quantile, shape = Quantile, fill = Quantile),
    size = 3, stroke = 0.8) +
```

```

# Median trend line
geom_smooth(
  data = filter(summ_age_long, Quantile == "Median"),
  aes(x = age, y = Expression),
  color = "black", linetype = "dashed",
  method = "loess", formula = y ~ x,
  linewidth = 1, se = FALSE
) +
# SCALES AND THEMING
scale_color_manual(
  values = color_map,
  breaks = c("Q25", "Median", "Q75"),
  labels = c("Q25 (25th percentile)", "Median", "Q75 (75th percentile)")
) +
scale_fill_manual(values = fill_map, na.value = NA,
  breaks = c("Q25", "Median", "Q75")) +
scale_shape_manual(values = shape_map, na.translate = FALSE,
  breaks = c("Q25", "Median", "Q75")) +
scale_x_continuous(breaks = seq(20, 90, 10)) +
scale_y_continuous(limits = c(0, 42), breaks = seq(0, 40, 5)) +
labs(
  title = "ABCA1 Expression by Age",
  x = "Age (years)",
  y = "ABCA1 Expression Level",
  caption = "Gray lines show Q25-Q75 range. Dashed line shows median trend."
) +
theme_bw(base_size = 16) +
theme(
  panel.grid.minor = element_blank(),
  panel.grid.major.x = element_blank(),
  plot.title = element_text(face = "bold", size = 18, hjust = 0.5),
  plot.subtitle = element_text(size = 12, hjust = 0.5, color = "gray40"),
  axis.title = element_text(face = "bold"),
  axis.text = element_text(color = "black"),
  legend.position = "right",
  legend.title = element_text(face = "bold"),
  legend.text = element_text(size = 12),
  plot.caption = element_text(color = "gray40", size = 10)
) +
guides(
  color = guide_legend(
    override.aes = list(
      shape = c(21, 21, 21),
      fill = c("#0072B2", "white", "#D55E00"),
      color = c("#0072B2", "black", "#D55E00"),
      linetype = c(0, 0, 0)
    )
  ),
  shape = "none",
  fill = "none"
)

# DISPLAY PLOT

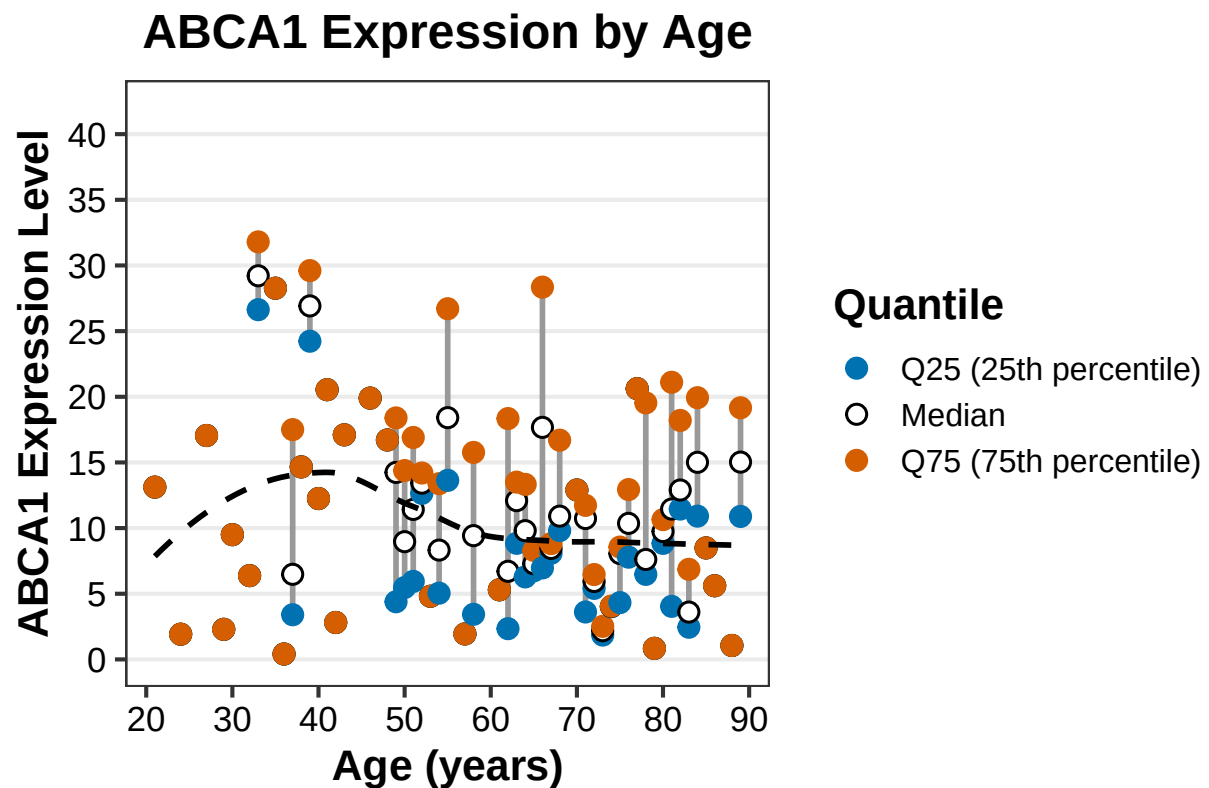
```

```
print(p_abca1)
```

```
## Warning: Removed 1 row containing non-finite outside the scale range  
## ('stat_smooth()').
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range  
## ('geom_segment()').
```

```
## Warning: Removed 3 rows containing missing values or values outside the scale range  
## ('geom_point()').
```



ray lines show Q25–Q75 range. Dashed line shows median trend.